Personality, Demographics and Polydrug Phenotypes: A Machine Learning Analysis of Co-Use Behaviors

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***Abstract*—** There have been several studies exploring the relationship between drug use, demographics, and personality traits. Substance use, in many cases, is a part of a wider polysubstance use. Yet, predictive modelling in this domain tends to treat each substance as a separate target. Using the UCI Drug Consumption (Quantified) dataset, this study tests how effective predicting drug habits by using demographic and personality traits really is. We approached the problem by replacing independent drug targets with clusters consisting of 3-4 different substances based on behavioral co-use. We created the groups by using Jaccard distance to capture co-occurrence patterns along with balanced k-medoids clusters. What we defined as active usage is the co-consumption of at least two drugs in a group within the last year (CL≥3). To avoid data leakage, we isolated a 30% split of the dataset to generate the clusters and used the remaining 70% with stratified cross-validation. We used a variety of models, including logistic regression, Extra Trees, Random Forest, HistGradientBoosting, XGBoost, and an MLP. The performance of the models reached a ceiling at about 0.75-0.82 balanced accuracy, with logistic regression being the top performer for most clusters. Interpreting our findings, it becomes clear that while demographic and personality factors are strong predictors, they fail to capture the full complexity of the drug co-use phenomenon, and more factors, such as environmental or social, may be needed to predict with more accuracy.

Keywords—Polysubstance use, Machine Learning, Class Imbalance, Phenotype Discovery, Cost-Sensitive Learning, Psychometrics

I. Introduction

According to the UNODC World Drug Report 2024, approximately 292 million people used drugs in 2022, which translates to about 1 in 18 people aged 15-64, and marks a 20% rise over the past decade [1]. Cannabis was the most commonly used substance [1]. What matters clinically is that use is often not confined to a single drug and follows polydrug patterns. Polydrug use is defined as "the concomitant or consecutive use of licit or illicit substances in a given period of time," according to the EMCDDA [2]. Using drugs can have negative effects on an individual's health, but polydrug use is linked to more severe outcomes. EUDA notes that consuming more than one substance in a short period of time results in more toxic interactions and increased risk of fatal and non-fatal overdoses, accidents, and hepatotoxicity [2]. EUDA's most recent analysis explains that overdose deaths rarely involve a single drug; the majority of these deaths involve the use of multiple substances, and detailed toxicology often detects multiple drugs at once [3].

Machine learning literature on drug usage often ignores the prevalence of co-use patterns and targets drug substances as separate, independent outcomes. This approach is often driven by the structure of open datasets like UCI Drug Consumption (Quantified), which labels each drug separately with a time-scale label ranging from CL0 ("never used") to CL6 ("used in the last day") [4]. In practice, these labels are often reduced to the binary target "user/non-user" with an adjusted use-frequency threshold [5], [6]. This approach comes with two significant limitations. Firstly, it does not capture the polydrug patterns emphasized in EUDA discussions [7] by treating drug use as an isolated event instead of naturally correlated behaviors. Secondly, the independent modeling of rare illicit drugs leads to class imbalance, as some drugs are used by a small sample of people. As a result of this, models can default to predicting only the majority class and are not able to identify actual users [8].

This motivated us to change what we ask our models to predict. Clinically, the goal was to detect the patterns that lead to high-risk co-use profiles since mixing drugs is often more dangerous than using a single substance alone [2], [7]. From a modeling standpoint, grouping rare illicit substances into broader phenotypes can solve the problem of data imbalance and increase the number of positive examples [8]. And because risk is not purely situational, we include psychometric measures as well, drawing on evidence that substance-use disorders are consistently associated with higher Neuroticism and lower Conscientiousness [9].

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In this study, we are shifting the modeling targets to reflect this clinical reality. We do not predict substances as independent and isolated targets. Instead, we focus on polysubstance phenotypes derived from observed co-use structures. We evaluate how demographic and psychometric variables relate to phenotype risk and use SHAP to understand which features have the most influence on the prediction outcomes [10].

II. Literature Review

In the field of addiction research, machine learning (ML) tends to rely heavily on supervised learning. This trend is confirmed by Mak et al. [11]'s systematic review. In the clinical reality, Acion et al. [12] compared standard statistical models with ML ensembles on a large national dataset to predict treatment success and demonstrated that ensembles achieved better results. Afshar et al. [13] used neural networks for multi-label detection to identify substance misuse in electronic health records, but the design was limited strictly to clinical texts and not to behavioral profiles.

As for survey-based research, the UCI Drug Consumption (Quantified) dataset [4] remains a widely utilized open benchmark. Fehrman et al. [6], who released the dataset, used multiple supervised classifiers to analyze drug-consumption risk using the Five-Factor Model (FFM) of personality, alongside some demographic features, sensation seeking, and impulsivity. The majority of their analysis was focused on a per-drug basis or via correlated groups. Notably, their analysis revealed that drug consumption follows a 'modular structure,' which they termed 'correlation pleiades' (e.g., a 'Heroin pleiad' correlating heroin, crack, and methadone). However, they discovered these groups by examining drug-to-drug statistical correlations and not by focusing on users' behavioral patterns. Furthermore, they did not attempt to predict polysubstance use; instead, they considered someone a user if they had used one of the drugs belonging to that "pleiad". This approach conflates single-substance use with high-risk co-use. Zoboroski et al. [14] used the same dataset but focused exclusively on predicting marijuana use, continuing this reductionist trend in drug prediction analysis. Through the comparison of logistic regression and neural networks, they identified important psychometric correlates of cannabis risk. However, by modeling marijuana alone, they overlooked the whole user's consumption history. This resulted in disconnecting the drug from co-use patterns that drive clinical risk [4], [14].

When treating substances independently, a major problem can emerge: class imbalance. The very few positive samples of rare illicit drugs in general population surveys often create very small positive classes compared to larger negative classes. As mentioned by Garcia and He [8], this leads standard classifiers to make biased predictions in favor of the majority class, inflating the accuracy numbers when, in reality, the models cannot efficiently distinguish a user from a non-user [8] [16]. Although we could use multi-label learning to see how drugs relate to one another [15], this approach is uncommon. The norm remains to predict substances individually, ignoring their relationships.

With all the above said, a clear gap remains in the literature. Even though Fehrman et al. [6] used statistical correlations between drugs to create "pleiades" they did not examine the possibility of creating clusters based on users' behavioral co-use patterns. This misses an opportunity to shift the focus from the drugs to the actual users. Specifically, we clustered drugs based on co-use patterns and assigned participants to these phenotypes using strict recency, which is closer to real-world behaviors and helps reduce the class imbalance problem [16].

III. Methodology

A. Dataset and Feature Selection

1) Dataset

The dataset we used to conduct our experiments is the UCI Drug Consumption (Quantified), created by Elaine Fehrman, Vincent Egan, and Evgeny Mirkes, and contains a total of 1885 responses [4], [6]. For each participant, demographic and psychometric information was extracted to create a profile. Then, they were asked to report drug usage with a time-scaled answer ranging from CL0 (never used) to CL6 (used in the last day) for every single one of the 19 substances, 18 valid substances, and one fake drug substance [4], [6]. Analytically, the substances, demographic, and psychometric information are demonstrated below [4]:

- Drugs: Alcohol, Amphetamines, Amyl, Benzodiazepines, Caffeine, Cannabis, Choc, Coke, Crack, Ecstasy, Heroin, Ketamine, Legal highs, LSD, Meth, Mushrooms, Nicotine, Semer, and VSA.
- Demographic: Age, Gender, Education, Country, Ethnicity
- Psychometrics: Ascore, Escore, Oscore, Cscore, Nscore, Impulsivity, Sensation Seeking (SS)

2) Feature Selection

Prior to any further analysis, we removed every dataset entry that reported "Semer" usage, a fictitious drug, to protect the data integrity and validity from people who overclaimed drug usage [4], [6]. Furthermore, we also excluded Caffeine and Choc, since these are daily-use substances with no illicit status. Including them would lead to their domination in the co-use structure, creating low specificity patterns with no actual meaning for our research.

B. *Leakage control via sequestered design*

Clustering is part of the target definition, so learning clusters on the full dataset can leak information from the evaluation split into the labels. To keep the evaluation clean, we implement a Sequestered Design [18]–[20]. We split the participants once using a fixed seed (RANDOM_STATE = 42). We reserve 30% of the available data as a design set to compute the similarities between drugs and create drug clusters. Once the clusters are created, we freeze them and use 70% of the data, where phenotype labels are instantiated using the created clusters, and all training and evaluation are performed, ensuring that phenotype definitions are learned without exposure to the evaluation sample [18]–[20].

C. *Drug–Drug Similarity and Polysubstance Phenotype Discovery*

The goal is to predict patterns of co-use. In order to do that, we first construct drug clusters that summarize which substances tend to be used by overlapping subsets of participants. All the steps done to discover phenotypes in this section are performed only on the design set (Section III-B) to avoid leakage [18]–[20].

1) Binarization for co-use structure

Each drug in the dataset is represented as a binary "used/not used" profile for every participant. Our goal at this stage is to discover phenotypes, and in order to achieve that, we discriminate users based on whether they have ever used the substance rather than focusing on when they did. For each participant u and each drug d, we define:

$$B_{u,d} = \mathbb{I}\left(CL_{u.d} \geq \tau_y\right), \tau_y = 1$$

where $\mathbb{I}(\cdot)$ is the indicator function. The result of this equation is a binary matrix $B \in \{0,1\}^{UxD}$. Each drug column $B_{\cdot,d}$ represents the set of participants who used the drug at least one time. As stated above, $\tau_y = 1$ is the threshold we used to discover phenotypes. We used a very loose limit to discover users in order to be able to discover patterns and drug-drug relationships more efficiently. Later on, we use a stricter limit in our models evaluation (Section III-D).

2) Drug-drug similarity using Jaccard distance

Once we had the binary profiles created, we needed a way to measure how "close" two drugs were behaviorally. We chose the Jaccard distance to achieve that because it solely focuses on the users, ignoring non-users [21]. In our sparse dataset, this is crucial, as most people are non-users to most of the drugs. If we tried to count "shared non-user" as a similarity, we would have the risk of every drug looking identical. To calculate the Jaccard distance for any two drugs d_j and d_i, we use the following formula:

$$\delta\left(d_i, d_j\right) = 1 - \frac{|U_i \cap U_j|}{|U_i \cup U_j|},$$

where $U_i = \{u: B_{u,i} = 1\}$ represents the group of participants who used the drug d_i. If we were to phrase this formula as a question, the question would be: "Out of everyone who used either drug, how many used both?". The result we get is a distance matrix that captures the co-use patterns [21].

3) Balanced k-medoids clustering

With the distances calculated, the next step was grouping drugs into K=4 distinct phenotypes. We chose to use a Balanced K-medoids algorithm rather than standard K-means. There were two reasons leading us to this choice. The first reason was that medoids are actual data points, which ensures each cluster is defined by a real drug rather than a synthetic average [22]. Second, standard clustering often creates one giant "junk" cluster and many small ones. This balance constraint ensures that there are no useless small clusters left, giving our models solid targets to work with [23].

To get the best result, we initialized the process using a "farthest-first" strategy [24] and ran the algorithm with 200 random restarts $N_{init} = 200$. We selected the solution with the lowest total cost and froze this final mapping $\{C1, \ldots, C4\}$ to serve as the ground truth for our predictive models [22]-[24].

D. *Phenotype label construction*

Having defined and frozen the clusters, we proceed to define the supervised targets using the remaining 70% modeling split of the dataset. In contrast to our previous approach for phenotype discovery, labeling adopts a stricter recency criterion to determine drug usage corresponding to last year's usage (CL3) [4], [6]. For every participant u and drug d, we define:

$$R_{u,d} = \mathbb{I}\left(CL_{u,d} \geq \tau_y\right), \tau_y = 3,$$

where $\mathbb{I}(\cdot)$ is the indicator function.

We then proceed to construct one binary target per cluster. A participant is considered a user if they have used at least two drug substances of the same cluster ($\mathcal{C}_k$) within the last year. This definition can be translated to the following mathematical function:

$$y_u^{\{(k)\}} = \mathbb{I}\left\{\sum_{\{d \in c_k\}} R_{u,d} \geq m\right\}, m = 2$$

This definition excludes single-substance activity from being treated as phenotype membership. To ensure evaluability, phenotype targets with extreme prevalence are removed; clusters are excluded when the positive class proportion on the modeling set falls below 1% or exceeds 99%.

E. *Handling Class Imbalance*

Most participants do not report illicit drug use; in many cases, this leads to our target phenotypes having far more negative cases (non-users) than positive ones. If we don't address this, the imbalance would cause our models to simply predict "No Use" for everyone to achieve high accuracy without actually learning anything [8].

In order to prevent this, we addressed the imbalance directly in the training process by using cost-sensitive learning instead of artificially resampling the data (e.g., SMOTE) [8], [16]. For the standard classifiers (Logistic Regression, Random Forest, Extra Trees), we applied class_weight= "balanced" [25]-[28]. This gives more weight to the errors made on the minority class, which discourages the model from tending to predict the majority class. For Gradient Boosting (XGBoost), we explicitly scaled the positive class weight using the ratio of negative to positive samples:

$$scaleposweigth = \frac{n_{neg}}{n_{pos}}$$

This makes mistakes in the minority class count more during training [29]. Moreover, we used stratified cross-validation to ensure that every fold has the same percentage (approximately) of users and non-users as the full dataset [30].

F. *Cluster-Level Interpretability via SHAP*

To interpret the Logistic Regression (LR) model's predictions—which is the best performing classifier as indicated in the Results section--we used SHAP (Shapley Additive Explanations) [10]. Beyond aggregate metrics (e.g., accuracy), SHAP explains how much each feature matters. For each participant, it calculates whether a certain trait increases or decreases the probability of them being classified as positive for a given phenotype [10]. To keep the analysis valid, these values were calculated on the Modeling split, making sure that the explanation phase remained separate from the clustering phase [18]-[20]. We ranked the features by averaging their absolute contribution across all participants, and created visual summaries presenting the results.

G. *Predictive models*

We transformed each phenotype into a binary classification task. We modeled several standard classifiers to evaluate their performance. Logistic Regression and MLP used standardized features, while tree-based models were trained on unscaled features [25]-[28], [31], [32]. Hyperparameters:

- *Logistic Regression: solver='liblinear', max_iter=2000, class_weight='balanced' [25], [28]. Preprocessing: StandardScaler [28].*
- Random Forest: n_estimators=400, min_samples_leaf=2, class_weight='balanced' [26], [28].
- Extra Trees: n_estimators=600, min_samples_leaf=2, class_weight='balanced' [27], [28].
- HistGradientBoosting: max_iter=300, max_depth=6, learning_rate=0.05 [28], [31].
- XGBoost: n_estimators=600, max_depth=4, learning_rate=0.05, subsample=0.85, colsample_bytree=0.85, reg_lambda=1.0. Imbalance handled via dynamic scale_pos_weight [29].
- MLP: Layers (64, 32), activation='relu', alpha=1e-4, max_iter=500, early_stopping=True [28], [32]. Preprocessing: StandardScaler [32].

H. *Evaluation Protocol and Metrics*

The performance was estimated on the modeling split using 5-fold stratified cross-validation (RANDOM_STATE = 42) [30]. The results reported were the mean score across folds. We didn't report plain accuracy and focused on metrics that remain informative when the positive class is rare, given the imbalance in some phenotype targets [8].

Balanced Accuracy (BA) is computed as the average of sensitivity and specificity [33]:

$$\mathrm{BA} = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right).$$

To summarize overall binary agreement under imbalance, we also report the Matthews Correlation Coefficient (MCC) [34]:

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}.$$

Minority-class retrieval is assessed using PR-AUC, reported as Average Precision (AP) over score thresholds [35], [36]:

$$\mathrm{AP} = \sum_{n} (R_n - R_{n-1}) P_n.$$

Finally, calibration is quantified with the Brier score (mean squared error between predicted probabilities p_i and labels y_i) [37]:

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N} (p_i - y_i)^2.$$

This is a short explanation of what each symbol means in the equations above:

BA -> Balanced Accuracy; TP -> True Positives; FN -> False Negatives; TN -> True Negatives; FP -> False Positives; MCC -> Matthews Correlation Coefficient; AP -> Average Precision; n -> Threshold/ranking index along the precision–recall curve; R_n -> Recall at index n; R_(n-1) -> Recall at index n-1; P_n -> Precision at index n; BS -> Brier Score; N -> Number of samples; i -> Sample index; p_i -> Predicted probability for sample i; y_i -> True label for sample i.

IV. Results

This section summarizes predictive performance for the four cluster-level phenotypes and compares the evaluated classifiers across clusters

A. Discovered Phenotypes

After removing all the participants answers who claimed to have used the fictitious drug Semer (8 participants removed), we had 1877 remaining responses in the dataset. After excluding caffeine/chocolate/Semer, and producing the four clusters with the procedure mentioned in Section III-C, the following groups were formed:

1. Cluster 1: Ketamine, Legal highs, LSD, Mushrooms
2. Cluster 2**:** Crack, Heroin, Meth, VSA
3. Cluster 3**:** Alcohol, Amyl, Cannabis, Nicotine
4. Cluster 4: Amphetamines, Benzos, Cocaine, Ecstasy

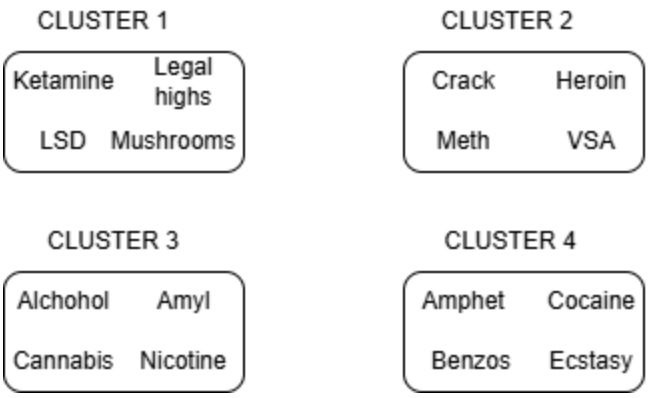

Fig. 1. Clusters discovered

There was significant variation in positive rates across the created clusters, ranging from 0.642 (Cluster 3) to 0.080 (Cluster 2). The low user-to-no-user ratio in cluster 2 made it the biggest challenge for our classifiers. In detail, the positive ratio for every cluster was:

1. Cluster 1: 0.260
2. Cluster 2: 0.080
3. Cluster 3: 0.642
4. Cluster 4: 0.307

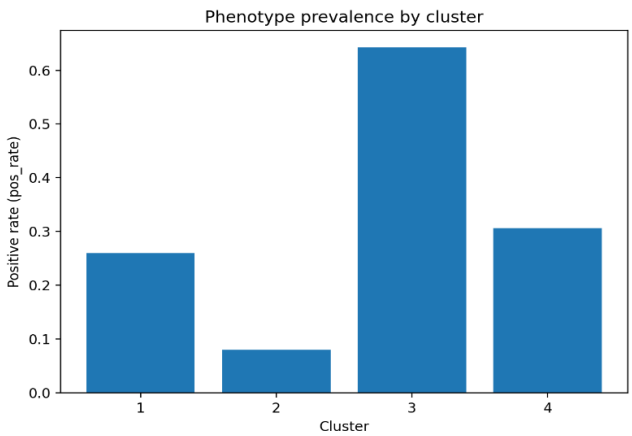

Fig. 2. Phenotype prevalence (positive rate) by cluster.

B. Overall Performance patterns across models

Across all the clusters, the best-performing model on discrimination-oriented metrics (balanced accuracy, MCC) was Logistic Regression (LR). While LR demonstrated stability, every other model could not reliably perform on cluster 2 due to its heavy imbalance. On the remaining three clusters, Random Forest and ExtraTrees managed to perform reliably, following LR closely. HistGradientBoosting, XGBoost, and MLP managed to be competitive but failed to surpass or follow the best three performing models closely. Overall, for all of our models, even for Logistic Regression, the task difficulty varied. Clusters 1,3,4 were manageable for all the models, while cluster 2 was a severe challenge for the majority of them.

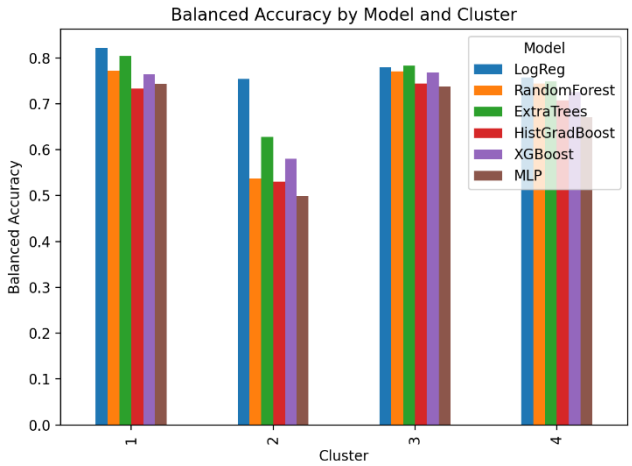

Fig. 3. Balanced Accuracy by Model and Cluster

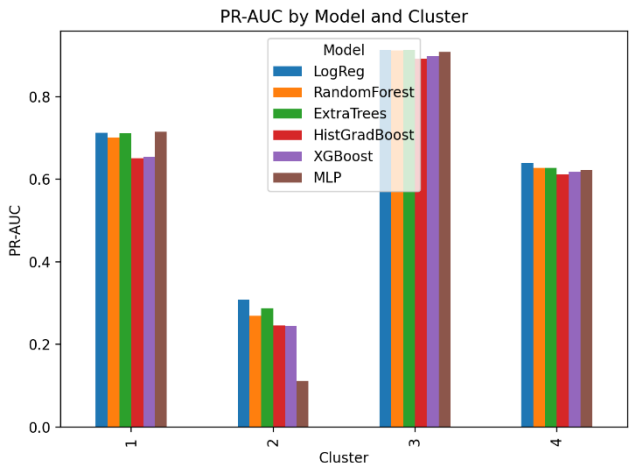

Fig. 4. PR-AUC by Model and Cluster

C. Per-Phenotype Predictive Performance

We evaluated the models by using the four metrics mentioned in Section III-H —Balanced Accuracy (BA), MCC, PR-AUC, and Brier score— in order to capture both calibration and discrimination. The results listed in the tables below are the average from 5-fold stratified cross-validation.

*1) **Cluster 1** (Ketamine, Legal Highs, LSD, Mushrooms; pos_rate = 0.260)*

Logistic Regression achieved the highest BA (0.822) and best overall agreement (MCC 0.589), which indicated strong discrimination between users and non-users. Extra Trees and Random Forest followed closely on both metrics. MLP reached a good level of PR-AUC but was not close to the three best models in the other two metrics. Brier was low on all the trained models, indicating good probability calibration.

Model	BA	MCC	PR-AUC	Brier
Logistic Regression	0.822	0.589	0.713	0.136
Random Forest	0.772	0.544	0.702	0.120
Extra Trees	0.804	0.574	0.711	0.123
HistGradientBoosting	0.733	0.475	0.651	0.141
XGBoost	0.764	0.515	0.655	0.144
MLP	0.743	0.523	0.716	0.122

Table 1. Cluster 1 performance. Mean over cross-validation folds.

*2) **Cluster 2** (Crack, Heroin, Meth, VSA; pos_rate = 0.080)*

Cluster 2 proved to be a fairly strong challenge for every model. This was mostly due to the high imbalance of the class with a low positive rate. Logistic Regression was the top-performing classifier, achieving the best BA (0.754) and the highest PR-AUC (0.309). Extra Trees was competitive on PR-AUC (0.287) and on MCC(0.250), but there was a notable difference between the two models in BA. MLP was the worst performer; it failed to learn under strong rarity. Brier score was very low for many models on this cluster, but this should be reviewed alongside PR-AUC and BA, in order to be interpreted correctly.

MODEL	BA	MCC	PR-AUC	Brier
Logistic Regression	0.754	0.300	0.309	0.168
Random Forest	0.536	0.109	0.270	0.069
Extra Trees	0.628	0.250	0.287	0.080
HistGradientBoosting	0.530	0.100	0.245	0.079
XGBoost	0.580	0.167	0.245	0.089
MLP	0.499	-0.009	0.111	0.129

Table 2. Cluster 2 performance. Mean over cross-validation folds

*3) **Cluster 3** (Alcohol, Amyl, Cannabis, Nicotine; pos_rate = 0.642)*

For the majority cluster, all our models showcased consistently high predictive performance. Extra Trees performed the best on BA(0.783), MCC(0.554), and PR-AUC(0.913), but with inconsiderable differences with Random Forest and Logistic Regression. The fact that many different models achieved good results with negligible differences indicates that predicting on this cluster was a straightforward task for our models given the current feature set.

MODEL	BA	MCC	PR-AUC	Brier
Logistic Regression	0.780	0.539	0.914	0.158
Random Forest	0.771	0.542	0.912	0.151
Extra Trees	0.783	0.554	0.913	0.152
HistGradientBoosting	0.744	0.494	0.892	0.171
XGBoost	0.769	0.528	0.899	0.169
MLP	0.737	0.492	0.909	0.156

Table 3. Cluster 3 performance. Mean over cross-validation folds

*4) **Cluster 4** (Amphetamines, Benzodiazepines, Cocaine, Ecstasy; pos_rate = 0.307)*

On Cluster 4, we had the second-highest but still moderate positive rate. Cluster 4 proved to be the second most challenging cluster for our classifiers. Logistic Regression was the top performer on this cluster, producing the best BA(0.757) and MCC(0.482). Random Forest and Extra Trees produced comparable results to Logistic Regression with small differences on BA, PR-AUC, MCC, and achieved better Brier Score, indicating higher probability calibration. HistGradientBoosting, MLP, and XGBoost could not outperform the three top models on any of the metrics.

Model	BA	MCC	PR-AUC	Brier
Logistic Regression	0.757	0.482	0.639	0.171
Random Forest	0.745	0.483	0.627	0.153
Extra Trees	0.748	0.472	0.628	0.157
HistGradientBoosting	0.708	0.421	0.611	0.171
XGBoost	0.725	0.438	0.618	0.175
MLP	0.671	0.375	0.622	0.164

Table 4. Cluster 4 performance. Mean over cross-validation folds.

D. Cluster-level Interpretation using SHAP on Logistic Regression

To understand the decision logic of our Logistic Regression baseline, we calculated SHAP values for each phenotype and summarized them using mean absolute SHAP magnitude per feature. Higher values on a feature indicate that the feature had a stronger influence on the model's output. These values explain what the model learned from the data, and they shouldn't be interpreted as relationships applying in the real world. For Cluster 1, the dominant factor was Age, followed by Gender/Country, then Oscore and SS. Cluster's 2 dominant features were country and gender, followed by Nscore and Age. In Cluster 3, Country and Age remain strong, with SS and Education following. Finally, in Cluster 4, the leading features are Country, SS, and Age, followed by Gender and Nscore. The SHAP feature plots following are more analytical, indicating the importance of each feature for every cluster.

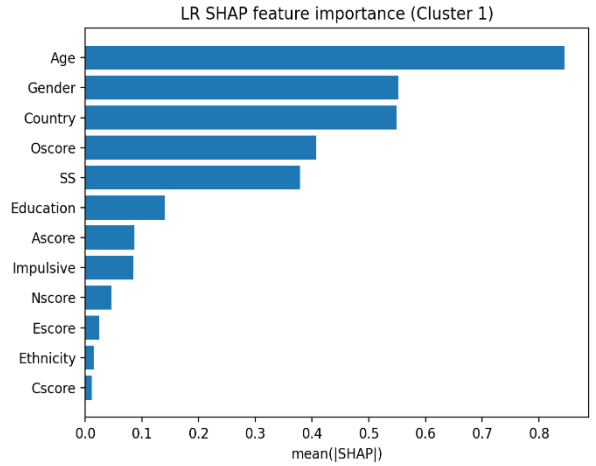

Fig. 5. LR SHAP feature importance for Cluster 1.

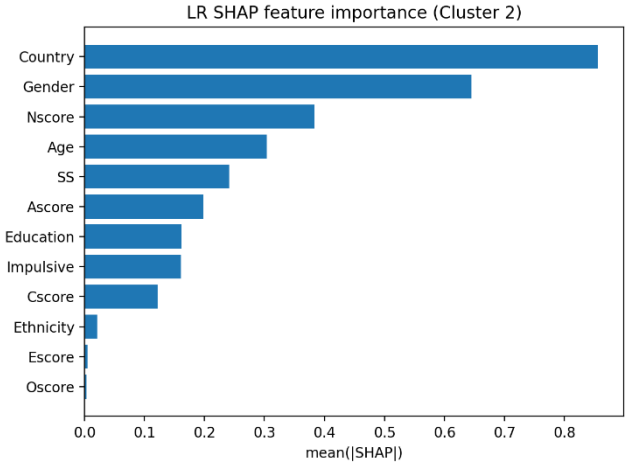

Fig. 6. LR SHAP feature importance for Cluster 2.

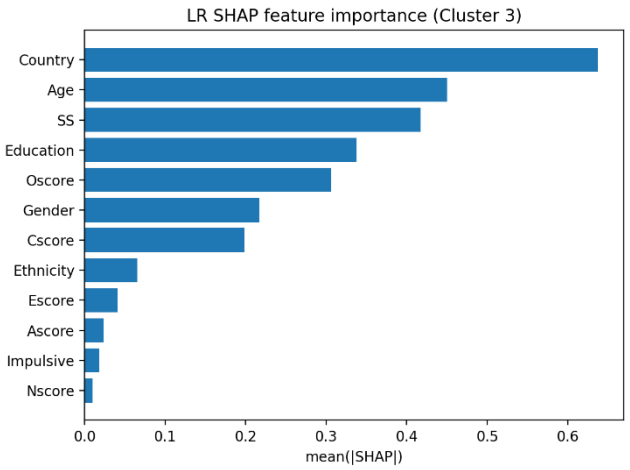

Fig. 7. LR SHAP feature importance for Cluster 3.

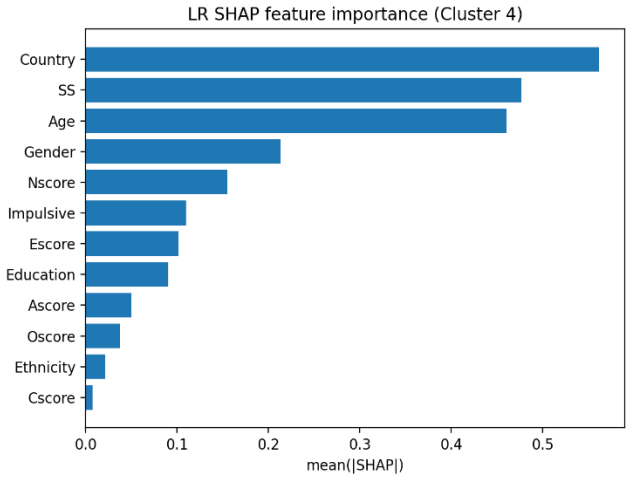

Fig. 8. LR SHAP feature importance for Cluster 4.

V. Discussion

This study's goal was to try a different approach to predicting substance use with machine learning. We changed the target from isolated drug substances to drug substance groups and tried to predict polydrug use [2], [7]. Someone was considered a user if they had used at least 2 substances belonging to a group within the last year. The groups were created by identifying behavior-based co-use phenotypes. Even though some studies proposed creating drug clusters based on drug-to-drug statistical correlations, we approached the clustering problem by shifting the focus to the user's behavior [6], [14].

Our results showed that although this approach was closer to clinical reality, it was not enough to solve the class imbalance problem present in general population surveys [8]. This was mostly present in the "hard drug" phenotype (Cluster 2), where the positive samples were just 8% of the total samples. Despite the application of cost-sensitive learning to penalize the majority class, even our best-performing models struggled to identify the users, with their PR-AUC peaking at 0.309 [35], [36]. This indicates that standard demographic and personality data do not provide enough information to give our models a good understanding of the patterns that lead to high-risk users in low-prevalence groups.

Apart from the class imbalance problem, another important finding needs to be mentioned. Even in our more balanced clusters, predictive accuracy seems to reach a ceiling at about 0.75-0.82 balanced accuracy, no matter which model was used [33]. Even more surprisingly, the more complex models, such as Extra Trees and XGBoost, rarely outperformed the simpler Logistic Regression baseline [27], [29]. Logistic regression was the top performer in three out of four clusters, presenting good results and good discrimination ability between users and non-users [28], [34]. This dominance also implies that the patterns distinguishing users are very direct, and using more complex models was unnecessary and ineffective.

The fact that all of our models hit a ceiling indicates that the problem is not the choice of algorithms, but the data features themselves. Although drug use and personality traits like Neuroticism are considered to be strongly connected with drug use [9], our model's prediction analysis using SHAP indicates that, in some clusters, demographics provided more information [10]. Surprisingly, in every cluster, three out of four most important features were demographics, and the most important feature for every cluster was either Country (Clusters 2, 3, 4) or Age (Cluster 1). By highlighting this, we don't mean to downplay the importance of personality on substance use interpretation, as SHAP proves that personality traits also provide useful information influencing our model's decisions [10]. Therefore, while demographic and personality features are important, they are not enough to fully explain the complexity of drug habits. Future studies need to include social or environmental factors, which would likely provide the extra information needed to achieve better and more accurate predictions [11], [17].

VI. Conclusion

In this study, we proposed a shift from modeling isolated substance use to predicting behavioral polysubstance phenotypes. While this approach is closer to clinical reality by capturing co-use patterns, the results indicated that clustering does not resolve the class imbalance problem. Severe class imbalance was present in the high-risk drug cluster (Cluster 2), which proved to be the most challenging for our classifiers. Among the trained models, Logistic Regression was the best performing model, with its balanced accuracy ranging from 0.75–0.82. Moreover, the performance ceiling of our models suggests that while demographic and psychological traits are strong predictors, they are not able to capture the full complexity of drug consumption habits. Future research must move beyond personal traits to integrate more environmental and social determinants, which could be useful to further enhance the prediction limits.

VII. References

[1] United Nations Office on Drugs and Crime (UNODC), *World Drug Report 2024: Special Points of Interest (SPI)*, 2024.

[2] European Union Drugs Agency (EUDA), "Polydrug use: health and social responses," *Mini-guide*, Oct. 2021.

[3] EUDA, "Drug-induced deaths – the current situation in Europe," in *European Drug Report 2025*, Jun. 2025.

[4] E. Fehrman, V. Egan, and E. Mirkes. "Drug Consumption (Quantified)," UCI Machine Learning Repository, 2015. [Online]. Available: https://doi.org/10.24432/C5TC7S.

[5] CRAN Package C443, "drugs: Drug consumption data set." [Online]. Available: https://rdrr.io/cran/C443/man/drugs.html

[6] E. Fehrman, A. K. Muhammad, E. M. Mirkes, V. Egan, and A. N. Gorban, "The Five Factor Model of personality and evaluation of drug consumption risk," *arXiv*:1506.06297, 2015.

[7] EMCDDA, *Polydrug use: patterns and responses*, Selected Issue, Nov. 2009.

[8] H. He and E. A. Garcia, "Learning from Imbalanced Data," in *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263-1284, Sept. 2009, doi: 10.1109/TKDE.2008.239.

[9] Kotov, R., Gamez, W., Schmidt, F., & Watson, D. (2010). Linking "big" personality traits to anxiety, depressive, and substance use disorders: A meta-analysis. *Psychological Bulletin, 136*(5), 768–821. https://doi.org/10.1037/a0020327

[10] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. (Also available as *arXiv*:1705.07874).

[11] Mak KK, Lee K, Park C. Applications of machine learning in addiction studies: A systematic review. Psychiatry Res. 2019 May;275:53-60. doi: 10.1016/j.psychres.2019.03.001. Epub 2019 Mar 4. PMID: 30878857.

[12] Acion, L., Kelmansky, D., Laan, M.D.V., Sahker, E., Jones, D. & Arndt, S. (2017) . Use of a machine learning framework to predict substance use disorder treatment success. *PLoS ONE*, *12*(4). http://dx.doi.org/10.1371/journal.pone.0175383

[13] Afshar, M., Sharma, B., Dligach, D., Oguss, M., Brown, R., Chhabra, N., Thompson, H. M., Markossian, T., Joyce, C., Churpek, M. M., & Karnik, N. S. (2022). Development and multimodal validation of a substance misuse algorithm for referral to treatment using artificial intelligence (SMART-AI): a retrospective deep learning study. *The Lancet Digital Health*, *4*(6), e426-e435. https://doi.org/10.1016/S2589-7500(22)00041-3

[14] Zoboroski, L., Wagner, T., & Langhals, B. (2021). Classical and Neural Network Machine Learning to Determine the Risk of Marijuana Use. *International Journal of Environmental Research and Public Health*, *18*(14), 7466. https://doi.org/10.3390/ijerph18147466

[15] M. -L. Zhang and Z. -H. Zhou, "A Review on Multi-Label Learning Algorithms," in *IEEE Transactions on Knowledge and Data Engineering*, vol. 26, no. 8, pp. 1819-1837, Aug. 2014, doi: 10.1109/TKDE.2013.39.

[16] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002. https://doi.org/10.48550/arXiv.1106.1813

[17] Negriff, Sonya & Dilkina, Bistra & Matai, Laksh & Rice, Eric. (2022). Using machine learning to determine the shared and unique risk factors for marijuana use among child-welfare versus community adolescents. PLOS ONE. 17. e0274998. 10.1371/journal.pone.0274998.

[18] Kapoor, Sayash & Narayanan, Arvind. (2023). Leakage and the reproducibility crisis in machine-learning-based science. Patterns. 4. 100804. 10.1016/j.patter.2023.100804.

[19] A. Moscovich and S. Rosset, "On the cross-validation bias due to unsupervised preprocessing," *J. R. Stat. Soc. Ser. B*, vol. 84, no. 4, pp. 1474–1502, 2022, doi: https://doi.org/10.48550/arXiv.1901.08974

[20] Apicella, Andrea & Isgrò, Francesco & Prevete, Roberto. (2025). Don't push the button! Exploring data leakage risks in machine learning and transfer learning.

Artificial Intelligence Review. 58. 10.1007/s10462-025-11326-3.
[21] Paul Jaccard. Étude comparative de la distribution florale dans une portion des Alpes et du Jura. 1901. DOI: 10.5169/SEALS-266450
[22] L. Kaufman and P. J. Rousseeuw, *Finding Groups in Data: An Introduction to Cluster Analysis*. New York, NY, USA: Wiley, 1990.
[23] Bernábe Loranca, María Beatríz & González Velázquez, Rogelio & Olivares-Benitez, Elias & Ruiz-Vanoye, Jorge & Flores,, José Luis. (2014). Extensions to K-Medoids with Balance Restrictions over the Cardinality of the Partitions. Journal of Applied Research and Technology. 12. 396-408. 10.1016/S1665-6423(14)71621-9.
[24] Gonzalez, T.F. (1985). Clustering to Minimize the Maximum Intercluster Distance. *Theor. Comput. Sci., 38*, 293-306.
[25] Nancy Reid FRS; Sir David R. Cox. 15 July 1924—18 January 2022. *Biogr. Mems Fell. R. Soc.* 1 December 2024; 77 69–91. https://doi.org/10.1098/rsbm.2023.0052
[26] Breiman, L. (2001). Random Forests. Machine Learning. 45. 5-32. 10.1023/A:1010950718922.
[27] Geurts, Pierre & Ernst, Damien & Wehenkel, Louis. (2006). Extremely Randomized Trees. Machine Learning. 63. 3-42. 10.1007/s10994-006-6226-1.
[28] Pedregosa, Fabian & Varoquaux, Gael & Gramfort, Alexandre & Michel, Vincent & Thirion, Bertrand & Grisel, Olivier & Blondel, Mathieu & Prettenhofer, Peter & Weiss, Ron & Dubourg, Vincent & Vanderplas, Jake & Passos, Alexandre & Cournapeau, David & Brucher, Matthieu & Perrot, Matthieu & Duchesnay, Edouard & Louppe, Gilles. (2012). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research. 12.
[29] Chen, Tianqi & Guestrin, Carlos. (2016). XGBoost: A Scalable Tree Boosting System. 785-794. 10.1145/2939672.2939785.
[30] Kohavi, Ron. (2001). A Study of Cross-Validation and Bootstrap for Accuracy Estimation and Model Selection. 14.
[31] Friedman, Jerome. (2000). Greedy Function Approximation: A Gradient Boosting Machine. The Annals of Statistics. 29. 10.1214/aos/1013203451.
[32] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, pp. 533–536, 1986, doi: 10.1038/323533a0.
[33] K. H. Brodersen, C. S. Ong, K. E. Stephan and J. M. Buhmann, "The Balanced Accuracy and Its Posterior Distribution," *2010 20th International Conference on Pattern Recognition*, Istanbul, Turkey, 2010, pp. 3121-3124, doi: 10.1109/ICPR.2010.764.
[34] Matthews BW. Comparison of the predicted and observed secondary structure of T4 phage lysozyme. Biochim Biophys Acta. 1975 Oct 20;405(2):442-51. doi: 10.1016/0005-2795(75)90109-9. PMID: 1180967.
[35] Saito, Takaya & Rehmsmeier, Marc. (2015). The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. PloS one. 10. e0118432. 10.1371/journal.pone.0118432.
[36] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," in *Proc. 23rd Int. Conf. Mach. Learn. (ICML)*, 2006, pp. 233–240, doi: 10.1145/1143844.1143874.
[37] G. W. Brier, "Verification of forecasts expressed in terms of probability," *Mon. Weather Rev.*, vol. 78, no. 1, pp. 1–3, 1950, doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.